

# A finite-difference method for solving the cubic Schrödinger equation

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## Abstract

A family of finite-difference methods is used to transform the initial/boundary-value problem associated with the nonlinear Schrödinger equation into a first-order, linear, initial-value problem. Numerical methods are developed by replacing the time and space derivatives by central-difference replacements. The resulting finite-difference methods are analysed for local truncation, errors, stability and convergence. The results of a number of numerical experiments are given.

## 1. Introduction

The nonlinear cubic Schrödinger equation (NLS) has the general form

$$i \frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial x^2} + q|u|^2 u = 0, \quad L_0 < x < L_1, \quad t > t_0, \quad (1)$$

where  $i = +\sqrt{-1}$ ,  $q \geq 0$  is a real parameter and  $u = u(x, t) = v(x, t) + iw(x, t)$ . This equation plays an important role in the modelling of several physical phenomena such as the propagation of optical pulses, waves in water, waves in plasmas and self-focussing effects in laser pulses. The complex-valued function  $u(x, t)$  governs the evolution of any weakly nonlinear, strongly dispersive, almost monochromatic wave (see [6, Chapter 17]).

Initial conditions associated with the partial differential equation (PDE) given in (1) will be assumed to have the form

$$u(x, t_0) = f(x) = f_R(x) + i f_I(x), \quad L_0 \leq x \leq L_1, \quad (2)$$

in which  $f_R$  and  $f_I$  are real-valued, continuous functions of  $x$ . Boundary conditions will be assumed to be of the form

$$\partial u(L_0, t)/\partial x = 0 \quad \text{and} \quad \partial u(L_1, t)/\partial x = 0, \quad t > t_0. \quad (3)$$

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The initial/ boundary-value problem (IBVP) (1)–(3) gives rise to soliton solutions in which the solution and its derivatives with respect to  $x$  vanish as  $|x| \rightarrow \infty$ . The function

$$u(x, t) = (2a/q)^{1/2} \exp[i\{\frac{1}{2}cx - (\frac{1}{4}c^2 - a)t\}] \operatorname{sech}[a^{1/2}(x - ct)] \quad (4)$$

with  $q \neq 0$  satisfies (1)–(3) [4]. For fixed  $t$ , this function decays exponentially as  $|x| \rightarrow \infty$ . It represents a soliton-type disturbance which travels with speed  $c$ ; its amplitude is governed by the real parameter  $a$ . In the case when  $q = 0$  in the NLS equation (1), the solution is given by

$$u(x, t) = \exp[i\{\frac{1}{2}cx - (\frac{1}{4}c^2 - a)t\}] \exp[a^{1/2}(x - ct)]. \quad (5)$$

Numerical methods for solving the IBVP (1)–(3) are also reported by Griffiths et al. [3] and Weideman and Herbst [5].

## 2. Numerical methods and their analyses

Following Bratsos [1], to obtain a numerical solution, the so-called method of lines will be used: this method transforms the IBVP (1)–(3) into a first-order initial-value problem (IVP). To this effect the region  $R = [L_0 < x < L_1] \times [t > t_0]$  with its boundary  $\partial R$  consisting of the ordinates  $x = L_0$ ,  $x = L_1$  and the axis  $t = t_0$ , is covered with a rectangular mesh,  $G$ , of points with coordinates  $(x, t) = (x_m, t_n) = (L_0 + mh, t_0 + nl)$  with  $m = 0, 1, \dots, N, N+1$  and  $n = 0, 1, \dots$ ; clearly  $h = (L_1 - L_0)/(N+1)$ . The solution of the NLS at the typical point  $(x_m, t_n)$  is, of course,  $u(x_m, t_n)$  which may be denoted, when convenient, by  $u_m^n$ . The solution of an approximating difference scheme at the same point will be denoted by  $U_m^n$ : for the purpose of analysing stability, the numerical value of  $U_m^n$  actually obtained (subject, for instance, to computer round-off errors) will be denoted by  $\tilde{U}_m^n$ . Collectively, the values  $U_m^n$  will be written in vector form as

$$U^n = [U_0^n, U_1^n, \dots, U_{N+1}^n]^T, \quad n = 0, 1, \dots, \quad (6)$$

$T$  denoting transpose. Clearly, there are  $N+2$  values to be determined at each time step.

The method of lines semi-discretizes the IBVP by replacing the space derivative in the NLS equation (1) by the familiar second-order central-difference approximant

$$\partial^2 u(x, t) / \partial x^2 = h^{-2} [u(x-h, t) - 2u(x, t) + u(x+h, t)] + O(h^2) \quad \text{as } h \rightarrow 0. \quad (7)$$

Eq. (1) with (7) is then applied to all  $N+2$  mesh points of  $G$  (see (6)) at time level  $t = t_n$  ( $n = 0, 1, \dots$ ). This leads to an IVP of the form

$$DU(t) = ih^{-2}AU(t) + iqBU(t), \quad t > 0; \quad U(t_0) = f, \quad (8)$$

where  $D = \operatorname{diag}\{d/dt\}$ ,  $f = [f_0, f_1, \dots, f_N, f_{N+1}]^T$  with  $f_m = f_R(x_m) + if_I(x_m)$  for  $m = 0, 1, \dots, N, N+1$ ,  $A$  is a tridiagonal matrix of order  $N+2$  given by

$$A = \begin{bmatrix} -2 & 2 & & & \\ 1 & -2 & 1 & & 0 \\ & \cdot & \cdot & \cdot & \\ 0 & & 1 & -2 & 1 \\ & & & 2 & -2 \end{bmatrix} \quad (9)$$

and

$$B = B(t) = \text{diag}\{|U_m(t)|^2\}. \quad (10)$$

Relation (8) gives

$$D = i(h^{-2}A + qB),$$

so

$$D^2 = -h^{-4}A^2 - qh^{-2}(BA + AB) - q^2B^2,$$

where  $A^2$  is a quindagonal matrix of order  $N + 2$  which is easily obtained from (9), and  $AB$  and  $BA$  are tridiagonal matrices of order  $N + 2$  which are easily obtained from (9) and (10).

Numerical methods will be developed by replacing the matrix-exponential term in the recurrence relation

$$U(t + l) = \exp(lD)U(t), \quad t = t_0 = 0, l, 2l, \dots, \quad (11)$$

where  $DU(t)$  is given by (8), by the family of Padé approximants to the exponential term in (11) given by

$$\exp(lD) = (I + \tilde{a}lD + \tilde{b}l^2D^2)^{-1}(I + \tilde{c}lD + \tilde{d}l^2D^2), \quad (12)$$

in which  $I$  is the identity matrix of order  $N + 2$  and the constants  $\tilde{a}$ ,  $\tilde{b}$ ,  $\tilde{c}$  and  $\tilde{d}$  are given in Table 1.

Substituting (12) in (11) gives

$$(I + \tilde{a}lD + \tilde{b}l^2D^2)U(t + l) = (I + \tilde{c}lD + \tilde{d}l^2D^2)U(t), \quad t = 0, l, 2l, \dots \quad (13)$$

Let  $p = l/h^2$ ,  $r = l/h$ ,  $f_1 = \tilde{a}p$ ,  $f_2 = \tilde{a}lq$ ,  $f_3 = \tilde{b}p^2$ ,  $f_4 = \tilde{b}qlp$  and  $f_5 = q^2\tilde{b}l^2$ . Similarly constants  $g_i$  ( $i = 1, \dots, 5$ ) can be defined for the right-hand side of (13). Then (13) subject to (8) and (10) gives a nonlinear algebraic system of the form

$$F^{n+1}(U^{n+1}) = G^n(U^n), \quad n = 0, 1. \quad (14)$$

The  $N + 2$  equations of this system are given in detail by

$$\begin{aligned} F_0^{n+1} = & U_0^{n+1} + 2if_1(-U_0^{n+1} + U_1^{n+1}) + if_2|U_0^{n+1}|^2 U_0^{n+1} - 2f_3(3U_0^{n+1} - 2U_1^{n+1} + U_2^{n+1}) \\ & - 2f_4(-|U_0^{n+1}|^2 U_0^{n+1} + |U_1^{n+1}|^2 U_1^{n+1}) - 2f_4|U_0^{n+1}|^2(-U_0^{n+1} + U_1^{n+1}) \end{aligned}$$

Table 1  
Parameters of eight Padé approximants to the exponential function

| Method | $\tilde{a}$ | $\tilde{b}$ | $\tilde{c}$ | $\tilde{d}$ |
|--------|-------------|-------------|-------------|-------------|
| (0,1)  | 0           | 0           | 1           | 0           |
| (1,0)  | -1          | 0           | 0           | 0           |
| (1,1)  | -1/2        | 0           | 1/2         | 0           |
| (1,2)  | -1/3        | 0           | 2/3         | 1/6         |
| (2,1)  | -2/3        | 1/6         | 1/3         | 0           |
| (0,2)  | 0           | 0           | 1           | 1/2         |
| (2,0)  | -1          | 1/2         | 0           | 0           |
| (2,2)  | -1/2        | 1/12        | 1/2         | 1/12        |

$$\begin{aligned}
& -f_5 |U_0^{n+1}|^4 U_0^{n+1}, \\
F_1^{n+1} &= U_1^{n+1} + i f_1 (U_0^{n+1} - 2U_1^{n+1} + U_2^{n+1}) + i f_2 |U_1^{n+1}|^2 U_1^{n+1} \\
& - f_3 (-4U_0^{n+1} + 5U_1^{n+1} - 4U_2^{n+1} + U_3^{n+1}) \\
& - f_4 (|U_0^{n+1}|^2 U_0^{n+1} - 2|U_1^{n+1}|^2 U_1^{n+1} + |U_2^{n+1}|^2 U_2^{n+1}) \\
& - f_4 |U_1^{n+1}|^2 (U_0^{n+1} - 2U_1^{n+1} + U_2^{n+1}) - f_5 |U_1^{n+1}|^4 U_1^{n+1}, \\
F_m^{n+1} &= U_m^{n+1} + i f_1 (U_{m-1}^{n+1} - 2U_m^{n+1} + U_{m+1}^{n+1}) + i f_2 |U_m^{n+1}|^2 U_m^{n+1} \\
& - f_3 (U_{m-2}^{n+1} - 4U_{m-1}^{n+1} + 6U_m^{n+1} - 4U_{m+1}^{n+1} + U_{m+2}^{n+1}) \\
& - f_4 (|U_{m-1}^{n+1}|^2 U_{m-1}^{n+1} - 2|U_m^{n+1}|^2 U_m^{n+1} + |U_{m+1}^{n+1}|^2 U_{m+1}^{n+1}) \\
& - f_4 |U_m^{n+1}|^2 (U_{m-1}^{n+1} - 2U_m^{n+1} + U_{m+1}^{n+1}) - f_5 |U_m^{n+1}|^4 U_m^{n+1}
\end{aligned} \tag{15}$$

for  $m = 2, 3, \dots, N-1$ ,

$$\begin{aligned}
F_N^{n+1} &= U_N^{n+1} + i f_1 (U_{N-1}^{n+1} - 2U_N^{n+1} + U_{N+1}^{n+1}) + i f_2 |U_N^{n+1}|^2 U_N^{n+1} \\
& - f_3 (U_{N-2}^{n+1} - 4U_{N-1}^{n+1} + 5U_N^{n+1} - 4U_{N+1}^{n+1}) \\
& - f_4 (|U_{N-1}^{n+1}|^2 U_{N-1}^{n+1} - 2|U_N^{n+1}|^2 U_N^{n+1} + |U_{N+1}^{n+1}|^2 U_{N+1}^{n+1}) \\
& - f_4 |U_N^{n+1}|^2 (U_{N-1}^{n+1} - 2U_N^{n+1} + U_{N+1}^{n+1}) - f_5 |U_N^{n+1}|^4 U_N^{n+1}
\end{aligned}$$

and

$$\begin{aligned}
F_{N+1}^{n+1} &= U_{N+1}^{n+1} + 2i f_1 (U_N^{n+1} - U_{N+1}^{n+1}) + i f_2 |U_{N+1}^{n+1}|^2 U_{N+1}^{n+1} \\
& - 2f_3 (U_{N-1}^{n+1} - 2U_N^{n+1} + 3U_{N+1}^{n+1}) - 2f_4 (|U_N^{n+1}|^2 U_N^{n+1} - |U_{N+1}^{n+1}|^2 U_{N+1}^{n+1}) \\
& - 2f_4 |U_{N+1}^{n+1}|^2 (U_N^{n+1} - U_{N+1}^{n+1}) - f_5 |U_{N+1}^{n+1}|^4 U_{N+1}^{n+1}.
\end{aligned}$$

Let, now,

$$Y^n = [V_0^n, V_1^n, \dots, V_{N+1}^n, W_0^n, W_1^n, \dots, W_{N+1}^n]^T, \tag{16}$$

then, using once more the expression  $u(x, t) = v(x, t) + iw(x, t)$  and equating the real and imaginary parts in each equation of the system (15) to zero, gives the nonlinear algebraic system

$$\tilde{F}^{n+1}(Y^{n+1}) = \tilde{G}^n(Y^n), \quad n = 0, 1, \dots, \tag{17}$$

which can be written in the form

$$\tilde{F}(Y^{n+1}) = \mathbf{0}, \quad n = 0, 1, \dots, \tag{18}$$

at each time step, where the vectors  $\tilde{F} = [\tilde{F}_0^{n+1}, \tilde{F}_1^{n+1}, \dots, \tilde{F}_{2N+4}^{n+1}]^T$ ,  $Y^{n+1}$  and the zero-vector  $\mathbf{0}$  are all of order  $2N + 4$ . The  $2N + 4$  equations of (18) are given, in detail, by

$$\begin{aligned}\tilde{F}_0^{n+1} &= V_0^{n+1} + 2f_1(W_0^{n+1} - W_1^{n+1}) - f_2|U_0^{n+1}|^2 W_0^{n+1} \\ &\quad - 2f_3(6V_0^{n+1} - 4V_1^{n+1} + 2V_2^{n+1}) - 2f_4(-|U_0^{n+1}|^2 V_0^{n+1} + |U_1^{n+1}|^2 V_1^{n+1}) \\ &\quad - 2f_4|U_0^{n+1}|^2(-V_0^{n+1} + V_1^{n+1}) - f_5|U_0^{n+1}|^4 V_0^{n+1}, \\ \tilde{F}_1^{n+1} &= V_1^{n+1} + f_1(-W_0^{n+1} + 2W_1^{n+1} - W_2^{n+1}) - f_2|U_1^{n+1}|^2 W_1^{n+1} \\ &\quad - f_3(-4V_0^{n+1} + 5V_1^{n+1} - 4V_2^{n+1} + V_3^{n+1}) \\ &\quad - f_4(|U_0^{n+1}|^2 V_0^{n+1} - 2|U_1^{n+1}|^2 V_1^{n+1} + |U_2^{n+1}|^2 V_2^{n+1}) \\ &\quad - f_4|U_1^{n+1}|^2(V_0^{n+1} - 2V_1^{n+1} + V_2^{n+1}) - f_5|U_1^{n+1}|^4 V_1^{n+1}, \\ \tilde{F}_m^{n+1} &= V_m^{n+1} + f_1(-W_{m-1}^{n+1} + 2W_m^{n+1} - W_{m+1}^{n+1}) \\ &\quad - f_2|U_m^{n+1}|^2 W_m^{n+1} - f_3(V_{m-2}^{n+1} - 4V_{m-1}^{n+1} + 6V_m^{n+1} - 4V_{m+1}^{n+1} + V_{m+2}^{n+1}) \\ &\quad - f_4(|U_{m-1}^{n+1}|^2 V_{m-1}^{n+1} - 2|U_m^{n+1}|^2 V_m^{n+1} + |U_{m+1}^{n+1}|^2 V_{m+1}^{n+1}) \\ &\quad - f_4|U_m^{n+1}|^2(V_{m-1}^{n+1} - 2V_m^{n+1} + V_{m+1}^{n+1}) - f_5|U_m^{n+1}|^4 V_m^{n+1}\end{aligned}$$

for  $m = 2, 3, \dots, N - 1$ ,

$$\begin{aligned}\tilde{F}_N^{n+1} &= V_N^{n+1} + f_1(-W_{N-1}^{n+1} + 2W_N^{n+1} - W_{N+1}^{n+1}) \\ &\quad - f_2|U_N^{n+1}|^2 W_N^{n+1} - f_3(V_{N-2}^{n+1} - V_{N-1}^{n+1} + 5V_N^{n+1} - 4V_{N+1}^{n+1}) \\ &\quad - f_4(|U_{N-1}^{n+1}|^2 V_{N-1}^{n+1} - 2|U_N^{n+1}|^2 V_N^{n+1} + |U_{N+1}^{n+1}|^2 V_{N+1}^{n+1}) \\ &\quad - f_4|U_N^{n+1}|^2(V_{N-1}^{n+1} - 2V_N^{n+1} + V_{N+1}^{n+1}) - f_5|U_N^{n+1}|^4 V_N^{n+1}, \\ \tilde{F}_{N+1}^{n+1} &= V_{N+1}^{n+1} + 2f_1(W_N^{n+1} - W_{N+1}^{n+1}) - f_2|U_{N+1}^{n+1}|^2 W_{N+1}^{n+1} \\ &\quad - f_3(2V_{N-1}^{n+1} - 4V_N^{n+1} + 6V_{N+1}^{n+1}) - 2f_4(|U_N^{n+1}|^2 V_N^{n+1} - |U_{N+1}^{n+1}|^2 V_{N+1}^{n+1}) \\ &\quad - 2f_4|U_{N+1}^{n+1}|^2(V_N^{n+1} - V_{N+1}^{n+1}) - f_5|U_{N+1}^{n+1}|^4 V_{N+1}^{n+1}, \\ \tilde{F}_{N+2}^{n+1} &= W_0^{n+1} + 2f_1(-V_0^{n+1} + V_1^{n+1}) + f_2|U_0^{n+1}|^2 V_0^{n+1} \\ &\quad - f_3(6W_0^{n+1} - 4W_1^{n+1} + 2W_2^{n+1}) - 2f_4(-|U_0^{n+1}|^2 W_0^{n+1} + |U_1^{n+1}|^2 W_{N+1}^{n+1})\end{aligned}$$

$$-2f_4 |U_0^{n+1}|^2 (-W_N^{n+1} + W_{N+1}^{n+1}) - f_5 |U_0^{n+1}|^4 W_0^{n+1},$$

$$\begin{aligned}\tilde{F}_{N+3}^{n+1} &= W_1^{n+1} + f_1 (V_0^{n+1} - 2V_1^{n+1} + V_2^{n+1}) \\ &\quad + f_2 |U_1^{n+1}|^2 V_1^{n+1} - f_3 (-4W_0^{n+1} + 5W_1^{n+1} - 4W_2^{n+1} + W_3^{n+1}) \\ &\quad - f_4 \left( |U_0^{n+1}|^2 W_0^{n+1} - 2|U_1^{n+1}|^2 W_1^{n+1} + |U_2^{n+1}|^2 W_2^{n+1} \right) \\ &\quad - f_4 |U_1^{n+1}|^2 (W_0^{n+1} - 2W_1^{n+1} + W_2^{n+1}) - f_5 |U_1^{n+1}|^4 W_1^{n+1},\end{aligned}$$

$$\begin{aligned}\tilde{F}_{N+1+m}^{n+1} &= W_m^{n+1} + f_1 (V_{m-1}^{n+1} - 2V_m^{n+1} + V_{m+1}^{n+1}) + f_2 |U_m^{n+1}|^2 V_m^{n+1} \\ &\quad - f_3 (W_{m-2}^{n+1} - 4W_{m-1}^{n+1} + 6W_m^{n+1} - 4W_{m+1}^{n+1} + W_{m+2}^{n+1}) \\ &\quad - f_4 \left( |U_{m-1}^{n+1}|^2 W_{m-1}^{n+1} - 2|U_m^{n+1}|^2 W_m^{n+1} + |U_{m+1}^{n+1}|^2 W_{m+1}^{n+1} \right) \\ &\quad - f_4 |U_m^{n+1}|^2 (W_{m-1}^{n+1} - 2W_m^{n+1} + W_{m+1}^{n+1}) - f_5 |U_m^{n+1}|^4 W_m^{n+1}\end{aligned}\quad (19)$$

for  $m = 2, 3, \dots, N-1$ ,

$$\begin{aligned}\tilde{F}_{2N+3}^{n+1} &= W_N^{n+1} + f_1 (V_{N-1}^{n+1} - 2V_N^{n+1} + V_{N+1}^{n+1}) + f_2 |U_N^{n+1}|^2 V_N^{n+1} \\ &\quad - f_3 (W_{N-2}^{n+1} - W_{N-1}^{n+1} + 5W_N^{n+1} - 4W_{N+1}^{n+1}) \\ &\quad - f_4 \left( |U_{N-1}^{n+1}|^2 W_{N-1}^{n+1} - 2|U_N^{n+1}|^2 W_N^{n+1} + |U_{N+1}^{n+1}|^2 W_{N+1}^{n+1} \right) \\ &\quad - f_4 |U_N^{n+1}|^2 (W_{N-1}^{n+1} - 2W_N^{n+1} + W_{N+1}^{n+1}) - f_5 |U_N^{n+1}|^4 W_N^{n+1}\end{aligned}$$

and

$$\begin{aligned}\tilde{F}_{2N+4}^{n+1} &= W_{N+1}^{n+1} + 2f_1 (V_N^{n+1} - V_{N+1}^{n+1}) + f_2 |U_{N+1}^{n+1}|^2 V_{N+1}^{n+1} - f_3 (2W_{N-1}^{n+1} - 4W_N^{n+1} + 6W_{N+1}^{n+1}) \\ &\quad - 2f_4 \left( |U_N^{n+1}|^2 W_N^{n+1} - |U_{N+1}^{n+1}|^2 W_{N+1}^{n+1} \right) - 2f_4 |U_{N+1}^{n+1}|^2 (W_N^{n+1} - W_{N+1}^{n+1}) \\ &\quad - f_5 |U_{N+1}^{n+1}|^4 W_{N+1}^{n+1}.\end{aligned}$$

The solution of the nonlinear system (18) can be determined either by fixed-point iteration or by Newton's method.

The local truncation error of all methods arising from (13) is given by

$$\begin{aligned}L(x, t) &= i(1 + \tilde{a} - \tilde{c}) \frac{\partial^2 u}{\partial x^2} + iq(1 + \tilde{a} - \tilde{c}) |u|^2 u + i(\tilde{a} - \tilde{c}) \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4} \\ &\quad + l \left\{ -(\tilde{b} - \tilde{d}) \left[ \frac{\partial^4 u}{\partial x^4} + q|u|^2 \frac{\partial^2 u}{\partial x^2} + q \frac{\partial(|u|^2 u)}{\partial x^2} + q|u|^4 u \right] \right\}\end{aligned}$$

$$+ \frac{1}{2} \frac{\partial^2 u}{\partial t^2} + \frac{\partial^3 u}{\partial x^2 \partial t} + \frac{\partial(|u|^2 u)}{\partial t} \Big\} + O(h^2 l) \quad \text{as } h, l \rightarrow 0, \quad (20)$$

from which it follows that, for consistency,  $\tilde{a}$  and  $\tilde{c}$  must be chosen so that  $1 + \tilde{a} - \tilde{c} = 0$ .

For the stability analysis of the methods, let  $U_s = \max |U_m^0|$  for  $m = 0, 1, \dots, N + 1$ . Then the linearization of the general term of the system (17) leads to the following stability equation:

$$\begin{aligned} & [1 + i f_2(U_s)^2 - f_5(U_s)^4] Z_m^{n+1} + [i f_1 - 2 f_4(U_s)^2] (Z_{m-1}^{n+1} - 2 Z_m^{n+1} + Z_{m+1}^{n+1}) \\ & - f_3(Z_{m-2}^{n+1} - 4 Z_{m-1}^{n+1} + 6 Z_m^{n+1} - 4 Z_{m+1}^{n+1} + Z_{m+2}^{n+1}) \\ & = [1 + i g_2(U_s)^2 - g_5(U_s)^4] Z_m^n + [i g_1 - 2 g_4(U_s)^2] (Z_{m-1}^n - 2 Z_m^n + Z_{m+1}^n) \\ & - g_3(Z_{m-2}^n - 4 Z_{m-1}^n + 6 Z_m^n - 4 Z_{m+1}^n + Z_{m+2}^n), \end{aligned} \quad (21)$$

where  $Z_m^n = U_m^n - \tilde{U}_m^n = e^{\alpha n l} e^{i \beta m h}$  in which  $\alpha$  is a complex number and  $\beta$  is real. Then substituting into (21) gives, after cancelling both sides by  $e^{\alpha n l} e^{i \beta m h}$ , the following stability equation:

$$\begin{aligned} & \left\{ [1 + i f_2(U_s)^2 - f_5(U_s)^4] - 4 [i f_1 - 2 f_4(U_s)^2] \sin^2 \frac{1}{2} \beta h - 16 f_3 \sin^4 \frac{1}{2} \beta h \right\} \xi \\ & = [1 + i g_2(U_s)^2 - g_5(U_s)^4] - 4 [i g_1 - 2 g_4(U_s)^2] \sin^2 \frac{1}{2} \beta h - 16 g_3 \sin^4 \frac{1}{2} \beta h, \end{aligned} \quad (22)$$

where  $\xi = e^{\alpha l}$  is the amplification factor. Eq.(22) is of the form  $A\xi = B$  with  $A, B \in \mathbb{C}$ , so that  $\xi = B/A$  when  $A \neq 0$ . Then the Von Neumann criterion for stability,  $|\xi| \leq 1$ , gives

$$\begin{aligned} & (\tilde{b} - \tilde{d}) \left[ q(U_s)^2 - \frac{1}{h^2} \sin^2 \frac{\beta h}{2} \right]^2 \left\{ 2 - l^2 (\tilde{b} + \tilde{d}) \left[ q(U_s)^2 - \frac{1}{h^2} \sin^2 \frac{\beta h}{2} \right]^2 \right\} \\ & \leq (\tilde{a}^2 - \tilde{c}^2) \left[ q(U_s)^2 - \frac{1}{h^2} \sin^2 \frac{\beta h}{2} \right]^2. \end{aligned} \quad (23)$$

Assuming that  $q(U_s)^2 - h^{-2} \sin^2 \frac{1}{2} \beta h \neq 0$ , the following condition for stability is derived from (23):

$$(\tilde{b} - \tilde{d}) \left\{ 2 - l^2 (\tilde{b} + \tilde{d}) \left[ q(U_s)^2 - \frac{1}{h^2} \sin^2 \frac{\beta h}{2} \right]^2 \right\} \leq \tilde{a}^2 - \tilde{c}^2. \quad (24)$$

It may be deduced from (24) that all the methods of the family (13) are stable, except those based on the (1,0) and (1,2) Padé approximants which are unstable.

For the convergence analysis consider the general equation of the system (19) and assume that  $U_m^n = e^{i n \psi} e^{i m \theta}$ , where  $\psi$  is a complex number and  $\theta$  is a real. Then, cancelling both sides by  $e^{i n \psi} e^{i m \theta}$  gives the following convergence equation:

$$\begin{aligned} & \left[ 1 + i \left( f_2 - 4 f_1 \sin^2 \frac{1}{2} \theta \right) - 16 f_3 \sin^4 \frac{1}{2} \theta + 8 f_4 \sin^2 \frac{1}{2} \theta - f_5 \right] e^{i \psi} \\ & = 1 + i \left( g_2 - 4 g_1 \sin^2 \frac{1}{2} \theta \right) - 16 g_3 \sin^4 \frac{1}{2} \theta + 8 g_4 \sin^2 \frac{1}{2} \theta - g_5. \end{aligned} \quad (25)$$

Since  $\theta$  is real, it follows from (25) that  $U_m^n$  remains bounded ( $\psi$  is real) whenever

$$(\tilde{b} - \tilde{d}) \left[ 2 - l^2 (\tilde{b} + \tilde{d}) \left( q - \frac{1}{h^2} \sin^2 \frac{\theta}{2} \right)^2 \right] \leq \tilde{a}^2 - \tilde{c}^2. \quad (26)$$

It may be deduced from (26) that all the methods arising from (13) are convergent, except those based on the (1,0) and (1,2) Padé approximants.

### 3. Numerical results

The IBVP (1)–(3) was solved numerically using a one-point iteration function to solve the nonlinear algebraic system at each time step. The boundary lines were taken to be  $L_0 = -80$  and  $L_1 = 100$  and the initial time was taken to be  $t_0 = 0$ . Taking  $a = 0.01$ ,  $c = 0.1$ ,  $q = 1.0$ , the solution modulus for  $0 \leq t \leq 108$  is depicted in Fig. 1. These values of  $a$ ,  $c$  and  $q$  were used to generate the initial distribution.

In computing numerical solutions, two stable numerical methods arising from (13) were used. These were obtained by substituting the following parameter values in (12):

$$\text{Method I} \quad a = -c = -\frac{1}{2}, \quad b = d = 0, \quad (27)$$

$$\text{Method II} \quad a = -c = -\frac{1}{2}, \quad b = d = \frac{1}{12}, \quad (28)$$

these are clearly the (1,1) and (2,2) Padé approximants to  $\exp(lD)$  in (11).

In a series of numerical experiments,  $h$  was given the values 2, 3, 5 and 6, while  $l$  was given the values 0.1, 0.2, 0.3 and 1.0. It was found that Method II, which arises from a higher-order approximation to  $\exp(lD)$  than Method I, did not produce an improvement in accuracy for any pair of values of  $h$  and  $l$ . This is explained by (20) which shows that the combinations in (27) and (28) produce local truncation errors with identical principal parts. A selection of these numerical results is reported in Table 2, where the error norm  $\|u - U\|_\infty$  is given, together with the value of  $x$  at which this maximum error modulus occurs. Starting with the (known) theoretical solution, the number,  $K$ , of iterations needed by a one-point iteration function to

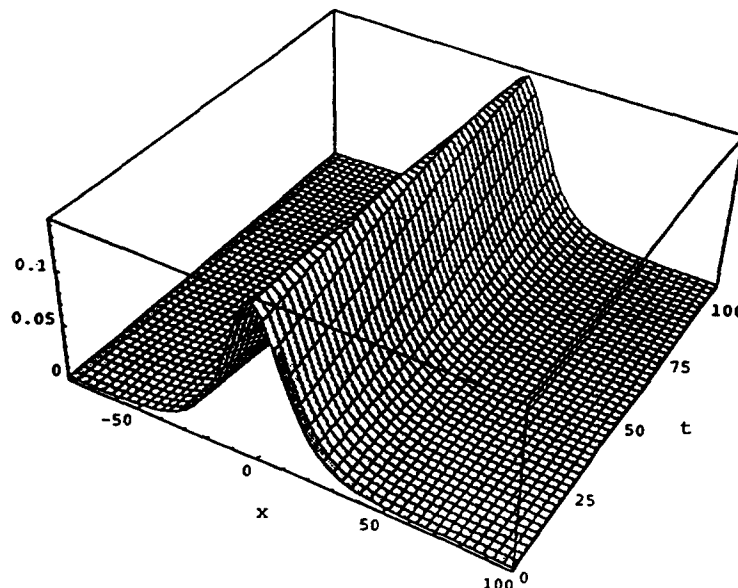


Fig. 1. Solution modulus for  $-80 \leq x \leq 100$ ,  $0 \leq t \leq 108$  with  $a_1 = 0.01$ ,  $c_1 = 0.1$ ,  $q = 1.0$ .



Table 2

Numerical results for Method I with  $a = 0.01$ ,  $c = 0.1$ ,  $q = 1$ 

| $h$ | $l$ | $t = 36$           |      |     | $t=108$            |      |     |
|-----|-----|--------------------|------|-----|--------------------|------|-----|
|     |     | $\ u - U\ _\infty$ | $x$  | $K$ | $\ u - U\ _\infty$ | $x$  | $K$ |
| 2   | 0.3 | 0.13E + 01         | – 70 | 14  | 0.15E + 01         | – 30 | 20  |
| 3   | 0.3 | 0.91E + 00         | – 62 | 6   | 0.95E + 00         | 97   | 8   |
| 5   | 0.1 | 0.64E – 01         | – 5  | 2   | 0.56E + 00         | 80   | 4   |
|     | 0.2 | 0.64E – 01         | – 5  | 3   | 0.56E + 00         | 80   | 4   |
|     | 0.3 | 0.64E – 01         | – 5  | 3   | 0.56E + 00         | 80   | 4   |
| 6   | 0.1 | 0.67E – 01         | – 8  | 2   | 0.39E + 00         | – 32 | 3   |
|     | 0.2 | 0.67E – 01         | – 8  | 3   | 0.39E + 00         | – 32 | 4   |
|     | 0.3 | 0.67E – 01         | – 8  | 3   | 0.39E + 00         | – 32 | 4   |
|     | 1.0 | 0.67E – 01         | – 8  | 3   | 0.40E + 00         | – 32 | 6   |

give two successive iterations (for all  $m$ ) agreeing with each other to four significant figures, is also given in Table 2.

In further experiments, Bratsos [1] varied  $a$ ,  $c$  and  $q$  while keeping  $h = 6$  and  $l = 0.3$  fixed. It was reported in [1] that for Method I with  $h$  and  $l$  fixed, especially for larger values of  $t$ ,

- (i)  $\|u - U\|_\infty$  increases as  $a$  is increased, with  $c$  and  $q$  fixed,
- (ii)  $\|u - U\|_\infty$  decreases as  $c$  is increased, with  $a$  and  $q$  fixed,
- (iii)  $\|u - U\|_\infty$  decreases as  $q$  is increased, with  $a$  and  $c$  fixed.

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